Guillaume Rachet

Lead Software Developer | Full-Stack Engineer (Angular, Scala, Java)
Specialized in Clinical Trial SaaS, Regulated Architecture & High-Performance Reporting



Merative / IBM Watson Health Dublin, Ireland | Oct 2021 – Present

Lead Full-Stack Software Engineer | June 2025 – Present

- **Architectural Transformation:** Directed the core reporting migration from legacy JSP/Oracle to an event-sourced Angular 19 and PostgreSQL JSONB model behind a unified REST API, retiring 10,000+ lines of debt and scaling to 3,000+ live reports for global enterprise clients.
- **Regulated Grammar & Logic Engine:** Engineered multi-select questions and set-logic operators (`INTERSECTS`, exact-set `==`) in ANTLR 4 across 12+ Scala and TypeScript subsystems (+3,500 test lines), unblocking major enterprise clinical trial contracts.
- **AI-Driven Leadership & Reliability:** Pioneered Claude Code and GitHub Copilot workflows to accelerate refactoring and TDD coverage while mentoring engineers and resolving critical GxP production escalations on rapid turnaround.

Senior Full-Stack Software Engineer | May 2024 – June 2025

- **Performance Engineering:** Slashed compliance and Audit Trail report execution times by 95% (150s to 7.5s), cut cold-cache data-summary latency on 200,000+ form studies from 74s to sub-second via O(1) lookups, and reduced monitoring query times by 72%.
- **Product Innovation (2x Award Winner):** Twice received the "Crushing It" Award for building the multi-sheet Excel Dashboard Summary and custom Excel-like column filtering across 16 core reports, becoming primary highlights in enterprise sales demos.
- **FDA Compliance & Async Batching:** Architected FDA-compliant Participant PDF batch generation (a key requirement for signing major CRO clients) and replaced synchronous export timeouts with an asynchronous background queue and polling workflow.

Full-Stack Software Engineer | Oct 2021 – May 2024

- **Modernized Identity & Access Management:** Led the migration of core authentication flows from legacy JSP to Angular/Scala across 25+ new REST endpoints supporting SSO, OTP, OTL, TOTP, and WAF bot protection, retiring 11,000+ lines of legacy code.
- **Enterprise Reporting & API Scaling:** Developed high-performance interactive report tables, role-validated scheduled email delivery, and consolidated 25 per-subject dashboard API calls into a single batch request to eliminate database bottlenecks.
- **Data Integration & SFTP Resilience:** Built the core ingestion UI and job orchestrator for the enterprise data migration module, and eliminated SFTP transfer hangs by engineering configurable connection timeouts and SSH keepalives.

Nokia Paris, France | Sept 2018 – Sept 2021

Software Engineer

- Engineered a greenfield React, Node.js, CouchDB, and WebSockets platform to automate courseware generation and file management for 100+ global technical authors.
- Integrated Microsoft Graph API for seamless SSO, automated email notifications, and dynamic SharePoint document compilation (PowerPoint, Word, Excel) directly in-app.
- Architected an internal QA review portal using React, TypeScript, GraphQL, and Material-UI connected to the corporate LMS to automate content grading and deployment workflows.

Dublin, Ireland (EU Citizen)
guillaume.rachet@gmail.com
+353 89 254 8122
linkedin.com/in/guillaume-rachet

Core Stack: Angular (v2–19, Signals, RxJS), Scala 2.13, TypeScript, Java, PostgreSQL (JSONB), React, Node.js
HTML5 / SCSS / JavaScript ES6+

Architecture & Leadership: Performance Engineering, Event Sourcing, Batch Pipelines, ANTLR 4, AI-Assisted Dev (Claude Opus / Copilot), REST APIs (JAX-RS)

Compliance & Testing: FDA 21 CFR Part 11, GxP Audit Trails, IAM (SSO/MFA/TOTP), TDD, ScalaTest, Vitest, JUnit 5

Tools & Ecosystem: Git, Docker, Carbon Design, Monaco Editor, OpenPDF, Apache POI, Jira, Figma, Agile/Scrum

Languages:
English (Fluent)
French (Native)

EDUCATION

Master's degree in Development & Software Engineering
AFORP - 2019 - 2021, Orsay France
NFQ Level 9 Equivalent

Bachelor's degree in Web/Mobile development
University Institutes of Technology Paris-Saclay - 2018 - 2019, Orsay France
NFQ Level 8 Equivalent

Associate degree in Computer Science
University Institutes of Technology Paris-Saclay - 2017 - 2018, Orsay France